

Smarter Sootblowing. Lower Steam Consumption.

Move from time-based cleaning to AI-driven, closed-loop sootblower automation for improved boiler efficiency and reliability.

20 TPD

Reduction in sootblowing steam consumption

~₹1 Crore / Year

Approximate annual savings at an internal steam cost of ₹1,500/tonne

29 sootblowers

Autonomously managed by the system

THE CHALLENGE

Your Sootblowers Are Cleaning. But Are They Cleaning When Needed?

Sootblowing is essential for removing ash and soot deposits from boiler heat-transfer surfaces. But there is a catch:

Too little cleaning → deposits build up and heat transfer suffers.

Too much cleaning → steam is wasted and tubes can experience unnecessary erosion.

In many plants, sootblowing still relies on:



Fixed time intervals

Blow every X hours



Predefined sequences

Follow the same sequence regardless of current conditions



Manual intervention

Operators decide when additional cleaning is required



Static logic

Limited ability to adapt to changing boiler conditions

The problem isn't whether the boiler is being cleaned. It's whether it is being cleaned at the right time, in the right place, and for the right reason.

THE SOLUTION

Faclon AI-Based Sootblower Automation

Faclon uses AI to continuously evaluate boiler operating conditions and determine when and where sootblowing is required.

01

Monitor

Continuously evaluate relevant boiler operating parameters.

02

Analyze

Identify changing conditions that indicate a cleaning requirement.

03

Decide

Determine the optimal sootblowing action instead of following a fixed schedule.

04

Execute

Automatically trigger sootblowers through the existing DCS infrastructure.

05

Adapt

Continuously adjust decisions based on changing boiler conditions.

On-premise deployment

OPC/DCS integration

Closed-loop automation

Fail-safe guardrails

THE SHIFT

From Conventional to Intelligent Sootblowing

CONVENTIONAL	FACLON AI-BASED
Fixed schedules Blow at predefined intervals	Dynamic decisions Determine when cleaning is actually required
Static sequences Follow predetermined blower sequences	Intelligent sequencing Select sootblowing actions based on current conditions
Manual intervention Operators monitor and intervene	Autonomous operation Execute decisions through the existing DCS
Time-based decisions Limited response to changing boiler conditions	Performance-driven Adapt to changing boiler conditions

Don't blow because it's time. Blow because the boiler needs it.

PROVEN IMPACT

20 TPD Less Sootblowing Steam

We recently completed a closed-loop AI implementation for sootblower automation on a 16 MW recovery boiler at a pulp mill.

29 sootblowers Autonomously managed by the system	DCS Integration Connected via OPC through a firewall	On-Premise Deployment Designed for controlled plant-floor operation
20 TPD Reduction in sootblowing steam consumption	~₹1 Crore / Year Approximate annual savings at an internal steam cost of ₹1,500/tonne	<i>Heat-transfer performance maintained during the measurement period.</i>

BEYOND STEAM SAVINGS

Intelligent sootblower automation can help plants move towards:

Lower auxiliary steam consumption Reduce unnecessary sootblower operation and conserve steam.	Reduced tube erosion Avoid excessive cleaning and support longer tube life.	Fewer boiler wash cycles Improve control of deposit buildup and reduce the need for intensive cleaning.
Reduced unplanned downtime More consistent boiler operation can help reduce operational disruptions.	Improved Net Heat Rate Better heat-transfer management can contribute to improved boiler performance.	Better flue-gas heat recovery Support more effective utilization of available heat.

Are your sootblowers running because the boiler needs cleaning, or because the schedule says they should? The difference can mean: steam saved, tubes protected, boiler performance maintained.

Ready to Make Sootblowing Intelligent?

Move from manual and time-based sootblowing to AI-driven, closed-loop automation.

👉 Book a Demo

👉 Explore AI-Based Sootblower Automation with Faclon

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